

Codec	Supported Bit Rates	Special Features	Downsides	High-Res Audio Support
SBC	up to 345kbps (theoretically)	Low power consumption Low processing power required"	Latency Issues Compressed audio	☐
AAC	up to 320kbps	Better compression than SBC	Not in open domain Uses more power and processing	☐
aptX	up to 384kbps	Better compression algorithm than SBC	Not supported on iOS	☒
aptX HD	up to 576kbps	Improvement over aptX Higher bit rate and bit depth than aptX	Not supported on iOS	☒
aptX LL	up to 352kbps	Low latency (about 32ms)	Not supported on smartphones	☐
LDAC	up to 990kbps	Supports High-Res Audio Better support for Android devices	Proprietary Sony technology Limited presence in the market	☒
ALAC	approx. 1411kbps	Supports lossless audio	Limited to Apple devices	☒

interested in the program, then I highly recommend you read this page – <https://digit.in/JackD>

06 Customise audio drivers

This tip is for the folks who have an affinity for coding and want to listen to awesome-quality music. Audio drivers that are supplied with your devices can often be limited to a certain intent of usage. While we do not recommend doing that with new audio hardware, if you have some old stuff lying around, ready to be experimented with, then go ahead. Creating custom drivers for your hardware would help you push your hardware that extra mile which you might have been looking for all this time. Here's a GitHub project that experiments with that, and I highly recommend you to have a look – <https://digit.in/CustDri>

07 Low-latency playback

One of the most important things in the audio space is the low-latency playback for devices. More often than not, the two devices that are communicating ensure that there is no noticeable lag. One of the first things that comes to mind is enabling low-latency mode on your device. Low-latency mode is something that is available on many TWS earbuds and is often advertised as game mode. Just keep in mind that turning low-latency mode on will cause your battery to drain faster. The other thing that you

can do is change the CODEC that is being used. Some CODECs, like Qualcomm's aptX promote low latency. Lastly, purist talk, whenever you can, use a wired connection. It's better.

08 Audio buffer sizes

If you are working with an advanced audio system, then one of the settings that I'll recommend you to change is the size of the buffer that is being used. Buffer size is the set amount of latency that is allowed during playback and recording. If you are changing this setting, then 512 samples or 11-12 milliseconds will provide optimal performance for your device. Now, depending on whether you want to play around with the stability of your system or not, you can increase or decrease the size of the buffer. For more in-depth information about buffer sizes, read this – <https://digit.in/BitDepth>

09 Implement upsampling

If you are familiar with the world of audio devices and players, you would know what sampling rate means. One of the best ways to elevate your audio listening experience is to upsample the audio. What happens here is identical bits of information called – zero-value samples – are inserted into the track. These days, most sources do not need upsampling. However, if you are someone who has been using an older system or are working with older audio samples, then its recommend.

10 Bit-perfect audio

When you are using audio playback software, it is normal for them to alter the outgoing signal by a bit (not in the literal sense). Now, if the factor of alteration is too high or the audio signal is being negatively impacted, you need to look into it. The easiest way to deal with this is to make sure that you have disabled all the digital signal processing options in your music player. Once that is out of the way, and you are still not getting the desired effect, then you need to look into alternatives for your music player. Some of the best hi-res audio players are listed here – <https://digit.in/HIRes>

11 Audio plugins

Audio plugins essentially are alterations to your output software that can expand the capabilities of the software that you are using. You can use plugins to add or block features of the software that you are using. Want to add a visualiser? Want to make your tracks sound like they are being sung by aliens from the 1990's? I am sure there will be a plugin for that and more. Just a simple Google search looking for the task you want to achieve along with the name of the software you are using to play music should lead you to the right way.

12 Customise audio presets

AWith music, in my and many other people's opinions, it is not like one